

Laundry Pickup System

Software Requirements

1. Chapter 1

1.1 System descriptions:

In today's fast-paced world, where convenience is highly valued, managing laundry services efficiently has become essential. The **Laundry Pickup System** makes it easier for customers to handle their laundry needs without leaving the comfort of their homes. The system offers a variety of features, such as creating and managing accounts, scheduling pickups and deliveries, selecting service types, and making secure online payments. Customers can also track the status of their orders in real-time and provide ratings or feedback once the service is completed.

The system enhances the overall laundry experience by ensuring smooth communication between customers, laundry providers, and delivery drivers. Laundry providers can efficiently manage orders, update statuses, handle pricing and services, assign deliveries, and generate detailed reports. Integration with external delivery providers ensures timely pickups, accurate delivery updates, and a seamless service flow.

1.2 The development Process model used:

The **Agile Development Model** is the most effective approach for developing the Laundry Pickup System. This method supports iterative and incremental development, allowing the team to deliver the system in manageable stages. Each phase focuses on implementing specific features, including account management, order creation, tracking, payment processing, delivery coordination, and reporting.

Using this model, the development team can gather feedback from users after each iteration and make necessary adjustments before proceeding to the next stage. This process ensures that the system continuously adapts to user needs and expectations, resulting in a **reliable, efficient, and user-friendly platform** for laundry service management.

1.3 Functional Requirements:

User Requirements

1. Customer shall be able to create an account.
2. Customer shall be able to log in.
3. Customer shall be able to edit their profile information.
4. Customer shall be able to specify location.
5. Customer shall be able to create a new laundry order.
6. Customer shall be able to track their order status.
7. Customer shall be able to make online payment.
8. Customer should be able to rate and provide feedback.
9. Laundry shall be able to view and manage customer orders.
10. Laundry shall be able to manage services and pricing.
11. Laundry shall be able to manage delivery requests.
12. Laundry shall be able to generate reports.
13. The system shall communicate with the external delivery provider.
14. The system shall update order delivery information.

System Requirements:

1. Customer shall be able to create an account.

- 1.1 The customer shall provide full name, phone number, email, and password.
- 1.2 The system shall verify the email and phone number for uniqueness.
- 1.3 The system shall confirm successful registration through a notification.

2. Customer shall be able to log in.

- 2.1 The customer must enter a valid email and password.
- 2.2 The system shall verify the credentials before granting access.
- 2.3 The system shall redirect the customer to their home page upon successful login.

3. Customer shall be able to edit their profile information.

- 3.1 The system shall provide an option to modify personal details such as name, phone, or address.
- 3.2 The customer shall confirm any changes before saving.
- 3.3 The system shall update and store the new information.

4. Customer shall be able to specify location.

- 4.1 The system shall detect the customer's location automatically using GPS.
- 4.2 The customer can manually enter or edit their address if needed.

5. Customer shall be able to create a new laundry order.

- 5.1 The customer shall select the type of service (Washing, Ironing, Washing & Ironing).
- 5.2 The customer shall choose preferred pickup and delivery times.
- 5.3 The system shall calculate and display the total cost.
- 5.4 The customer shall confirm the order before submission.
- 5.5 The system shall send the delivery request to the external delivery provider.

6. Customer shall be able to track their order status.

- 6.1 The system shall display the current order status (Picked Up, In Progress, Ready, Delivered).
- 6.2 The system shall show delivery updates received from the external delivery provider.

7. Customer shall be able to make online payment.

- 7.1 The system shall support online payment using Mada or Apple Pay.
- 7.2 The system shall display a confirmation message once the payment is successful.

8. Customer should be able to rate and provide feedback.

- 8.1 The system should allow customers to rate their experience after order completion.
- 8.2 The system should allow customers to write comments or feedback about the laundry service

9. Laundry shall be able to view and manage customer orders.

- 9.1 The system shall display all pending, in-progress, and completed orders.
- 9.2 The laundry shall be able to accept or reject new customer orders.
- 9.3 The laundry shall be able to update each order's status (In Progress, Ready, Delivered).

10. Laundry shall be able to manage services and pricing.

- 10.1 The laundry shall be able to add, update, or remove services.
- 10.2 The laundry shall be able to modify the price of each service.
- 10.3 The system shall automatically display updated prices to customers.

11. Laundry shall be able to manage delivery requests.

11.1 The system shall send delivery requests to the external delivery provider once an order is confirmed.

11.2 The system shall update the order status when delivery updates are received from the provider.

12. Laundry shall be able to generate reports.

12.1 The system shall provide daily and monthly order reports.

12.2 The system shall display revenue summaries and performance statistics.

13. The system shall communicate with the external delivery provider.

13.1 The system shall send order details (pickup address, delivery address, and times) to the provider.

13.2 The system shall receive delivery status updates (Picked Up, Delivered) from the provider.

14. The system shall update order delivery information.

14.1 The system shall display delivery progress to the customer based on the updates received.

14.2 The system shall mark the order as “Delivered” once confirmation is received from the provider.

1.4 Non-Functional Requirements:

User Requirements:

1. The system will ensure user data security.
2. The system will support all major operating systems for smooth functionality.
3. The system will be compatible with all major web browsers.
4. The system will provide an interface that is easy to use.
5. The system will be able to handle an increasing number of users.
6. The system will be available for users at all times.

System Requirements:

1. The system will ensure user data security

- 1.1 The system must encrypt all customer, driver, and laundry data using AES-256 encryption.
- 1.2 The system must implement secure authentication using OTP or email verification.
- 1.3 The system must apply role-based access control for customers, drivers, and laundry admins.
- 1.4 The system must log all user actions for auditing and monitoring purposes.

2. The system will support all major operating systems for smooth functionality.

- 2.1 The system must run seamlessly on all major mobile and desktop operating systems.
- 2.2 The system must provide consistent performance while supporting at least 1,000 concurrent users per operating system.
- 2.3 The system must be developed using Flutter for the front-end and Node.js for the back-end to ensure cross-platform compatibility.

3. The system will be compatible with all major web browsers.

- 3.1 The system must function properly on Chrome, Firefox, Safari, and Edge, and be tested on at least 3 different versions of each browser to ensure compatibility.
- 3.2 The system must maintain layout consistency and responsive design across all browsers, and be tested on at least 5 different screen resolutions to ensure proper display.
- 3.3 The system must support at least 500 simultaneous web sessions without performance degradation or errors.

4. The system will provide an interface that is easy to use.

- 4.1 The system must have a clear and simple navigation structure.
- 4.2 The system must display all buttons, icons, and menus clearly.
- 4.3 The system must provide feedback for user actions, such as order submission or updates.
- 4.4 The system must include help tips or guidance for complex tasks.

5. The system will be able to handle an increasing number of users.

- 5.1 The system must support at least 1,500 concurrent users without performance degradation.
- 5.2 The system must efficiently manage high loads during peak times, such as weekends or holidays.
- 5.3 The system must allow for future integration of additional laundries, drivers, or services.

6. The system will be available for users at all times.

- 6.1 The system must achieve at least 99.9% uptime.
- 6.2 The system must switch automatically to backup servers in case of failure.
- 6.3 The system must perform regular database backups every 12 hours.